Engine Model(s): PT6A

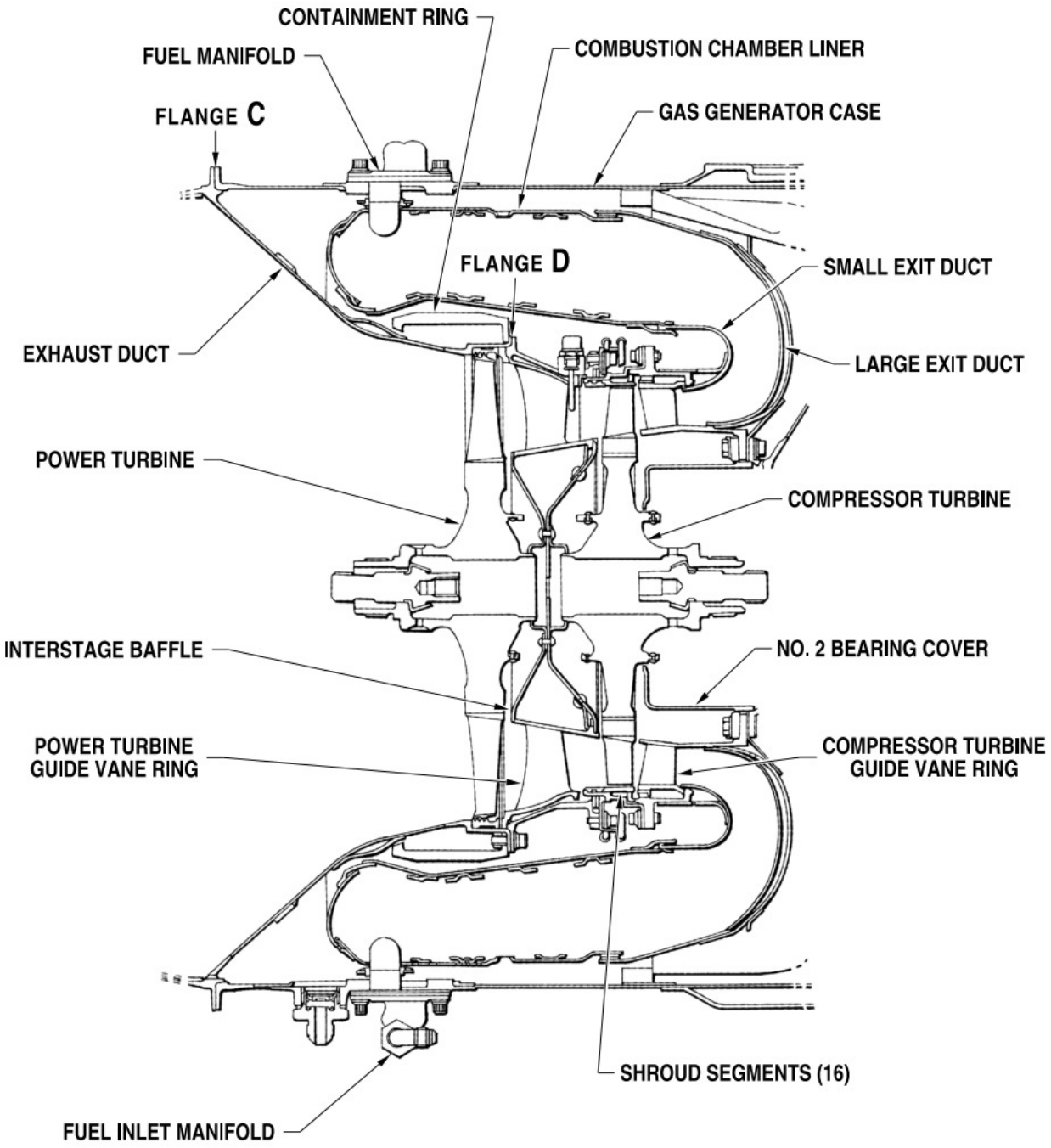
72-50-00

TURBINES SECTION - DESCRIPTION AND OPERATION

1. General (Ref. Fig. 1)

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Figure 1 Compressor Turbine and Power Turbine - Cross-section



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The engine incorporates two turbines, one for the compressor section and the other for the power section. The compressor turbine is splined to the compressor front stubshaft, while the power turbine is splined to the power turbine shaft assembly, which, in turn, drives the reduction gearbox (Ref. 72-10-00).

Each turbine is preceded by an inlet guide vane assembly which directs the gas flow onto its associated turbine at the most efficient angle. The compressor turbine inlet guide vanes are provided with cored cooling air passages (Ref. 75-30-00). The compressor turbine rotates within a shroud housing which incorporates 16 shroud segments. The segments are ground to provide the necessary blade tip clearance.

The power turbine vane ring and interstage baffle, which form part of the power turbine stator assembly, are contained within the power turbine stator housing which is bolted at its front end, together with power turbine containment ring, to the exhaust duct.

The gas flow from the combustion chamber liner is directed into the engine turbine area via the annular passage between the small and large exit ducts. This passage changes the gas flow 180 degrees to provide a forward flow.

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